

Hermes Kanban

Multi-Agent Profile Collaboration for Hermes Agent

*A durable, profile-aware work-queue architecture for coordinating
heterogeneous agent workloads across research, operations, and engineering*

Hermes & Teknium
Nous Research

April 25, 2026 | Design Spec, Revision 01

Abstract

This document specifies **Hermes Kanban**, a multi-agent collaboration layer for the Hermes Agent framework. It defines a durable, SQLite-backed task board shared across named agent *profiles*, allowing heterogeneous agents—researchers, editors, engineers, persistent digital-twin assistants—to collaborate asynchronously on long-lived work without the lifecycle fragility of in-process subagent swarms.

We compare the proposed architecture against three contemporary systems: Cline Kanban’s worktree-per-task board, Paperclip’s persistent-agent company orchestrator, and NanoClaw’s failed in-process agent-team SDK. We further contextualize the design against Google’s April 2026 Gemini Enterprise Agent Platform, which splits the agentic stack between a governance-oriented control plane and a velocity-oriented execution plane. Hermes Kanban deliberately stays at the execution plane, with governance expressible as user-space profiles and plugins rather than kernel features.

The specification covers: a minimal SQLite schema (tasks, links, comments, events), a dumb cron-driven dispatcher with atomic claims, eight canonical collaboration patterns, four user stories spanning research / scheduled operations / persistent digital twins / coding, a first-class orchestrator profile pattern (addressing the common “my orchestrator does the work itself” failure mode), an optional single-column multi-tenant extension that lets one specialist fleet serve many business contexts, and an implementation plan organized as eight independently shippable pull requests. The kernel requires no changes to `run_agent.py`, no new core tools, and no tool-schema expansion on any API call.

*Status: DESIGN ONLY. No implementation accompanies this document.
Target audience: Hermes maintainers and contributors; design-review participants.*

Contents

1	Motivation	4
1.1	The multi-agent gap in Hermes today	4
1.2	What others have built, and what broke	4
1.3	Google Gemini Enterprise: control vs. execution	5
1.4	Design thesis	5
2	Related Systems: Comparative Analysis	7
3	Architecture	8
3.1	Overview	8
3.2	Three-plane separation	8
3.3	The critical invariant: no in-process subagent swarms	8
4	Data Model	9
4.1	SQLite schema	9
4.2	Status semantics	10
4.3	Workspace kinds	10
5	Collaboration Patterns	11
5.1	P1 — Fan-out	11
5.2	P2 — Pipeline	11
5.3	P3 — Voting / Quorum (fan-in)	11
5.4	P4 — Long-running journal	11
5.5	P5 — Human-in-the-loop triage	11
5.6	P6 — @mention delegation	11
5.7	P7 — Thread-scoped workspace	11
5.8	P8 — Fleet farming	12
6	The Orchestrator Profile	13
6.1	Three properties of a well-behaved orchestrator	13
6.2	Shipping implication: installable profile templates	13
6.3	Why this is not kernel work	14
7	Multi-Tenant Context	15
7.1	The tenant primitive	15
7.2	What “tenant” actually means	15
7.3	What stays shared across tenants	15
7.4	What is deliberately <i>not</i> here	15
7.5	The “small company serving multiple clients” ask, solved	16
8	Worked Example: 50-Account Social Media Fleet	17
8.1	Setup	17
8.2	Per-tick task generation	17
8.3	The dispatcher does what it does	17
8.4	Why this is clean	18
9	User Stories	19
9.1	Story 1: Research triage and synthesis (non-coding)	19
9.2	Story 2: Scheduled recurring workflow (non-coding)	20
9.3	Story 3: Digital-twin / persistent assistant role (non-coding)	20
9.4	Story 4: Coding pipeline (code-shaped)	20

9.5 What all four stories share	20
10 Kanban vs. <code>delegate_task</code>	22
11 Assignment Semantics	23
12 Dispatcher Design	24
12.1 Concurrency correctness	24
13 CLI and Gateway Surface	25
13.1 Command surface	26
13.2 Gateway integration	27
14 Scope Boundaries	28
15 Implementation Plan	29
15.1 Files touched	29
16 Risks, Tradeoffs, Open Questions	31
16.1 Risks	31
16.2 Tradeoffs	31
16.3 Open questions	31
17 Conclusion	32